

The Architecture of Cloud Foundry

This article explains the architecture of Cloud Foundry, a cloud virtualisation platform that is gaining popularity.



So what is a cloud virtualisation platform? It is a layer of software that sits between the software application that you write, and the cloud platform. Cloud virtualisation platforms help us avoid the ‘Hotel California’ problem of vendor lock-in: “*You can check out anytime you like, but you can never leave!*” Once you write your application using a particular cloud vendor’s API, it will be very difficult for you to move to another platform, which is, essentially what vendor lock-in is all about. Cloud virtualisation allows us to avoid precisely this situation. Examples of cloud virtualisation platforms are Google App Engine and Salesforce Heroku. The Cloud Foundry platform, available at cloudfoundry.org, is Apache 2.0-licensed, which is a very friendly open source licence. So if you’re interested in knowing what else sets Cloud Foundry apart, it is:

- Supported on many cloud platforms: Amazon, Rackspace, OpenStack, Ubuntu, etc.
- Supports many programming languages and frameworks: Java, Scala, Ruby, Spring, Rails, Grails, etc.
- Supports many back-end services: RabbitMQ, MySQL, PostgreSQL, MongoDB, Redis, etc.

The history of Cloud Foundry

VMware developed Cloud Foundry and contributed it to the open source community. Being open source is expected to increase the acceptability of the platform and make it widely accessible. It is located at www.cloudfoundry.org. The source code can be obtained from <http://github.com/cloudfoundry>. Note that the cloudfoundry.org site is different from <http://www.cloudfoundry.com/>, which is where VMware provides a hosted and commercially supported version of Cloud Foundry.

Architecture

Let’s look at the major components of the Cloud Foundry platform, which are written in Ruby, and run as daemons.

The Cloud Controller

This component of the architecture orchestrates all other components. It also stores information about the status of other components in the architecture, the users, the deployed applications and available services. It exposes a REST interface for accepting requests from the Cloud Foundry